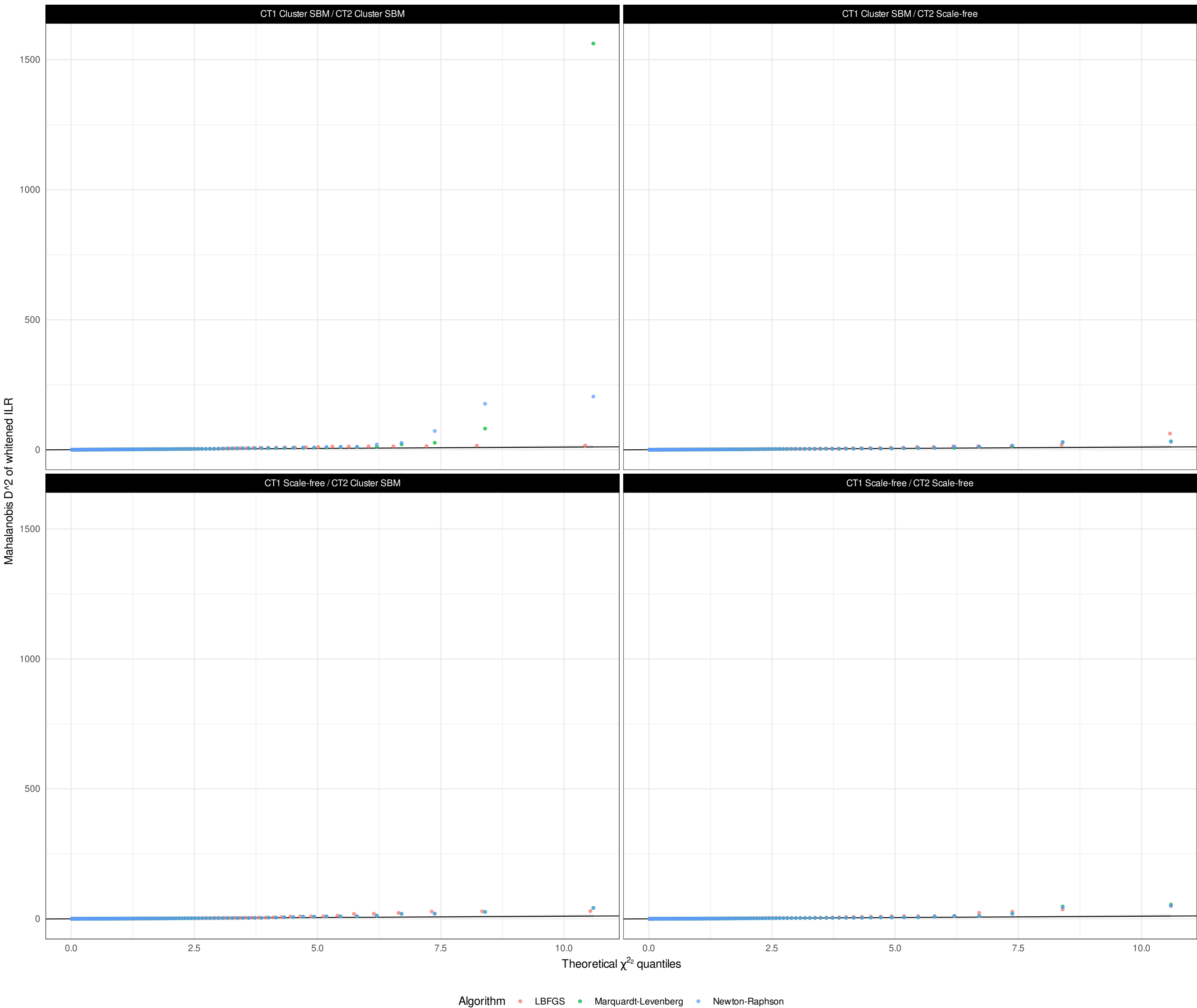
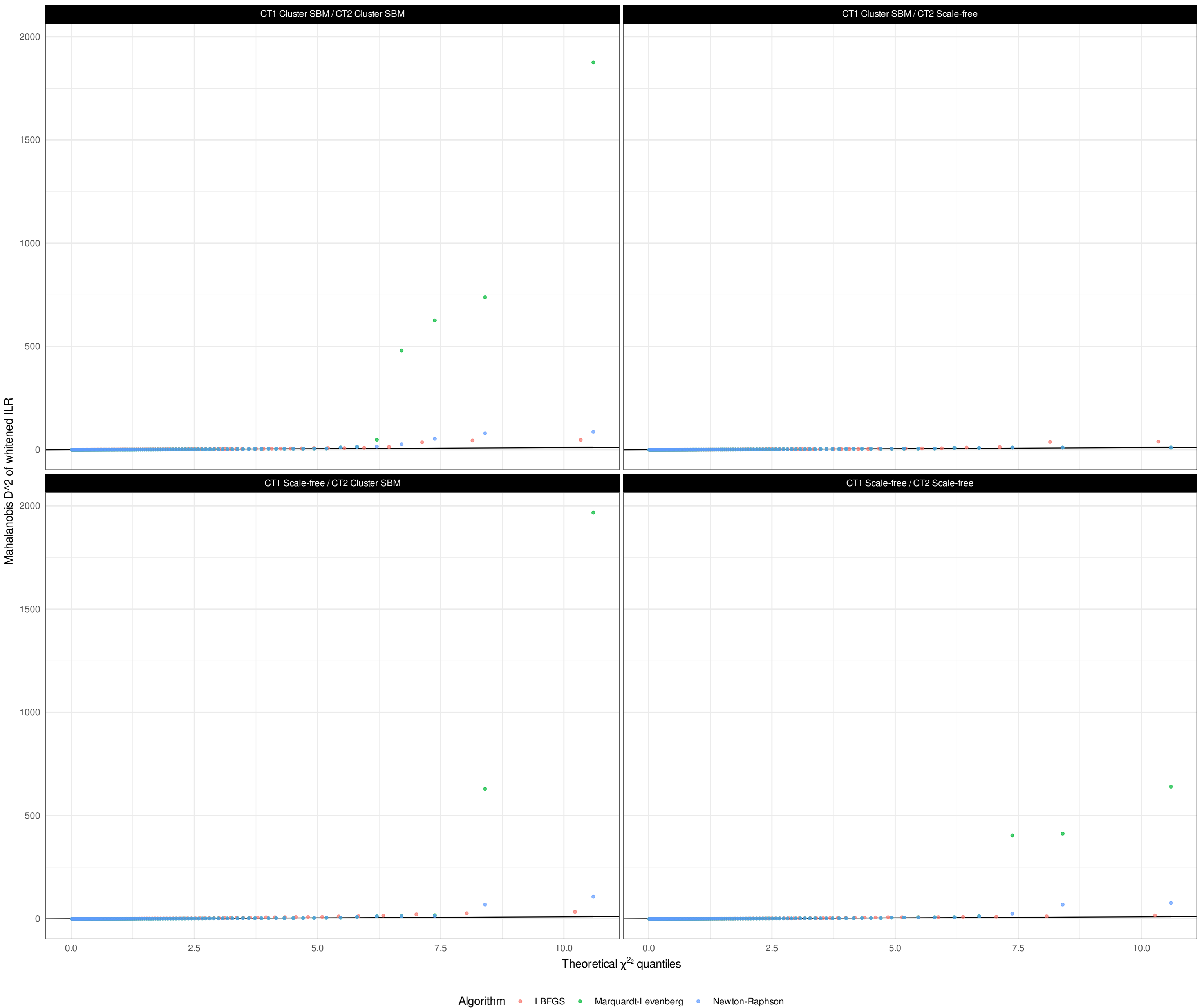
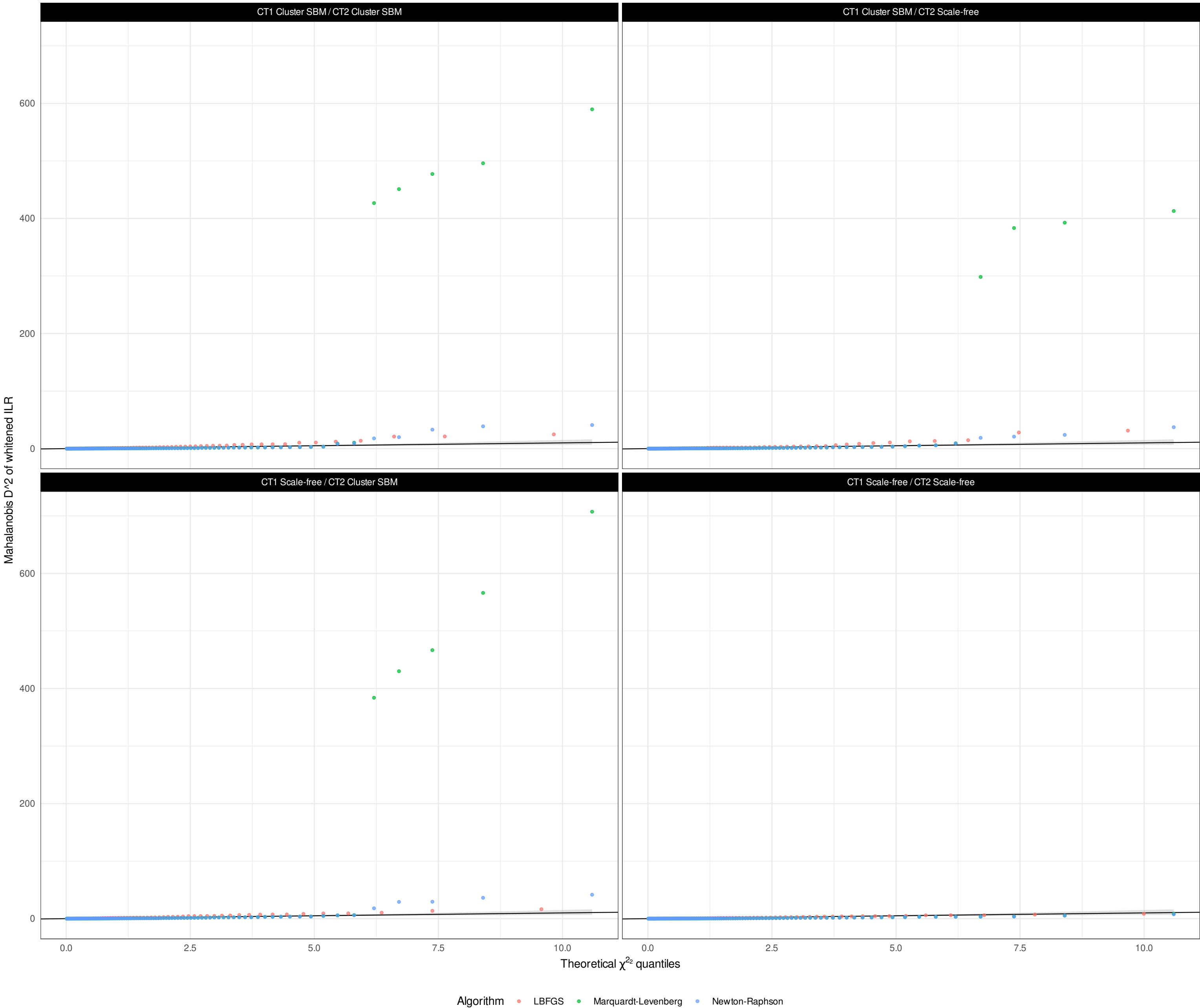
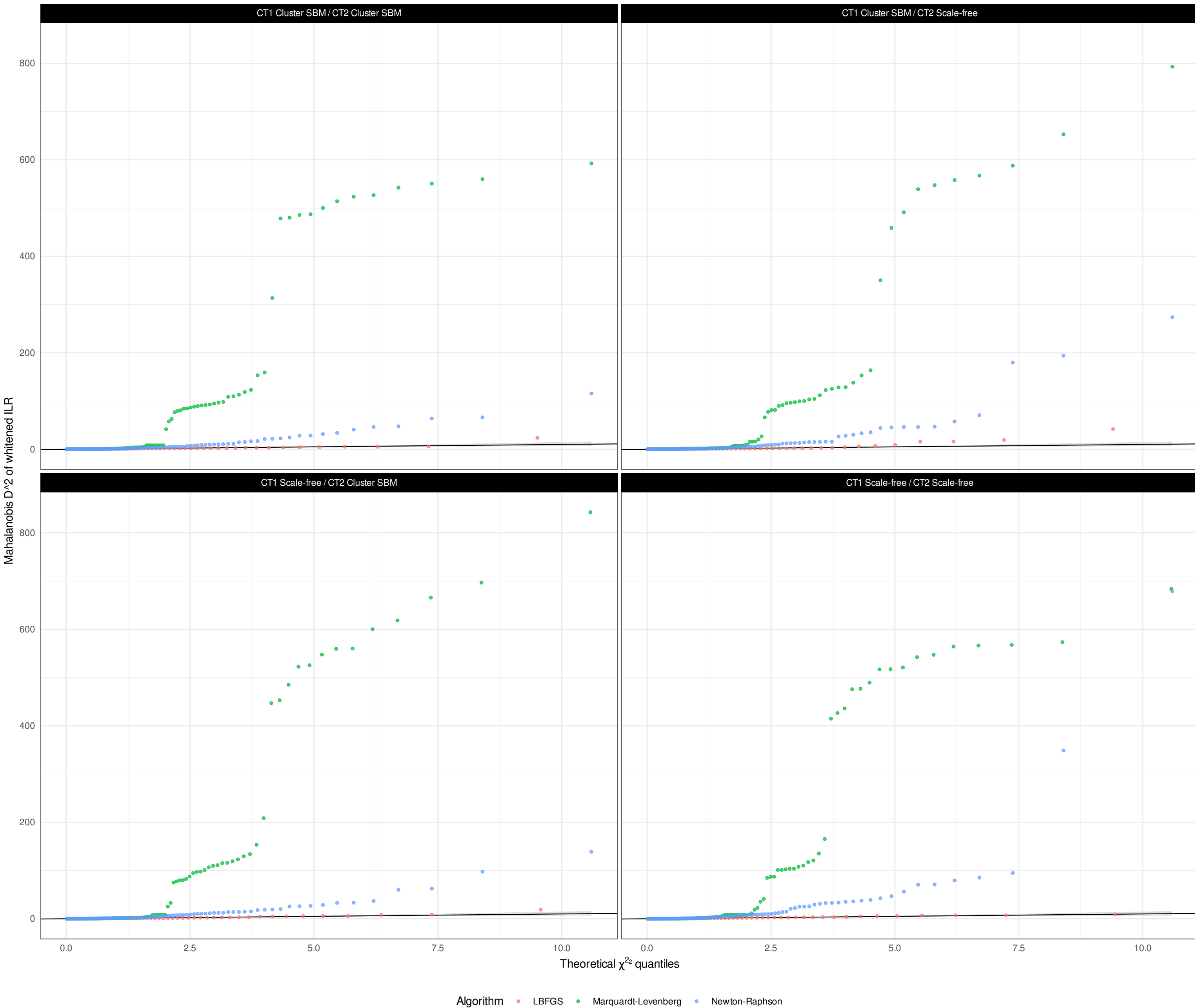


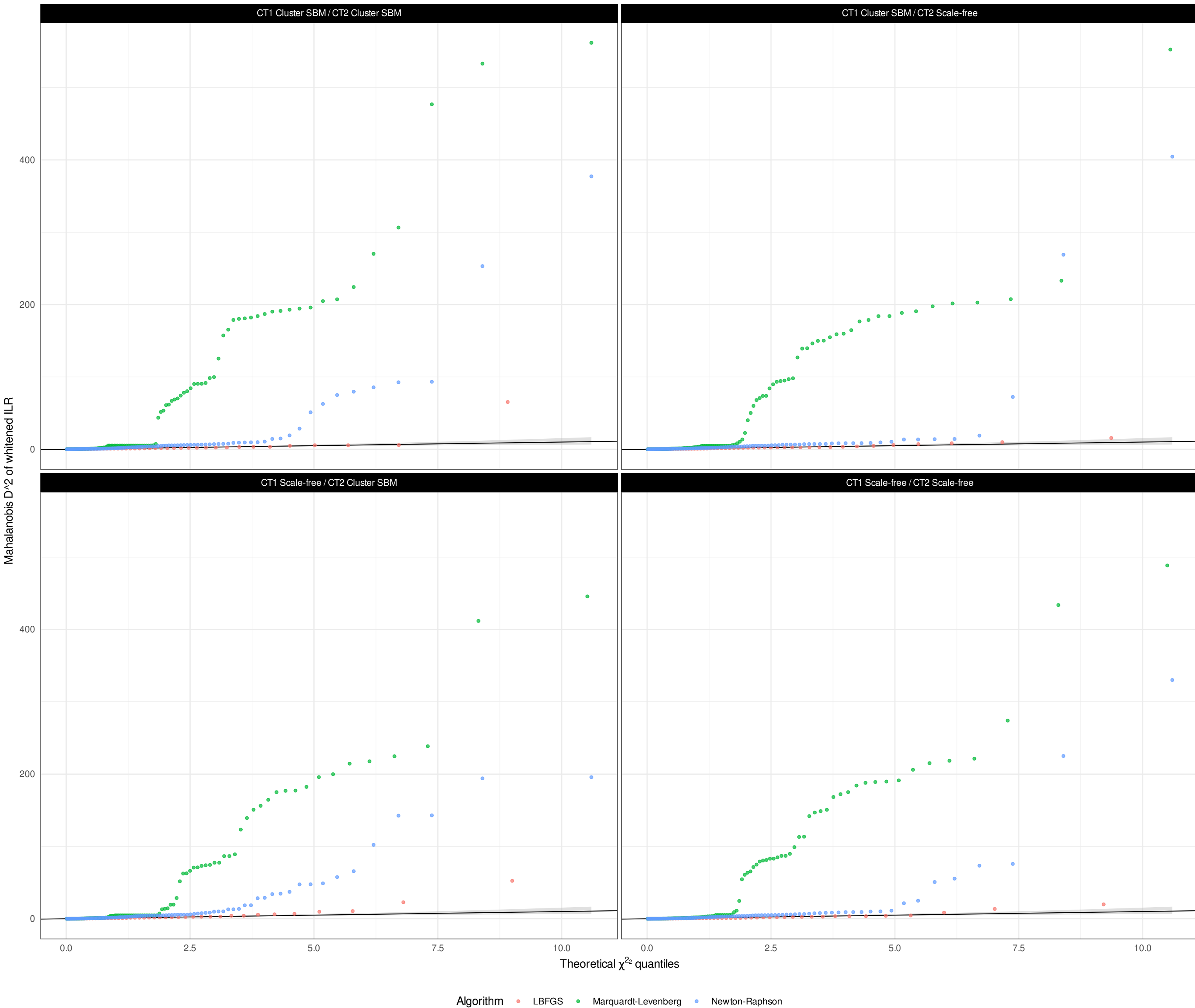






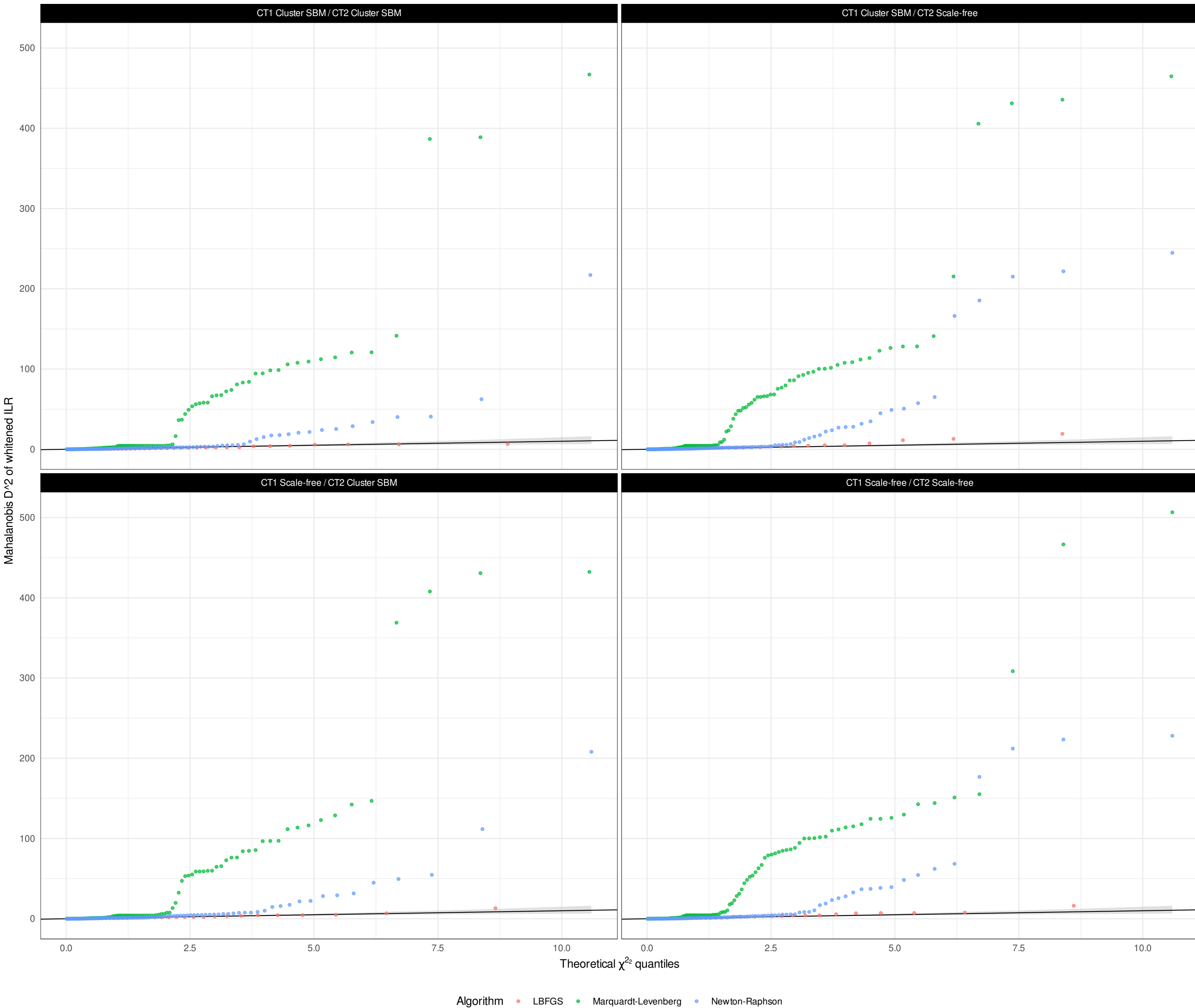


moderately unbalanced / moderate OVL



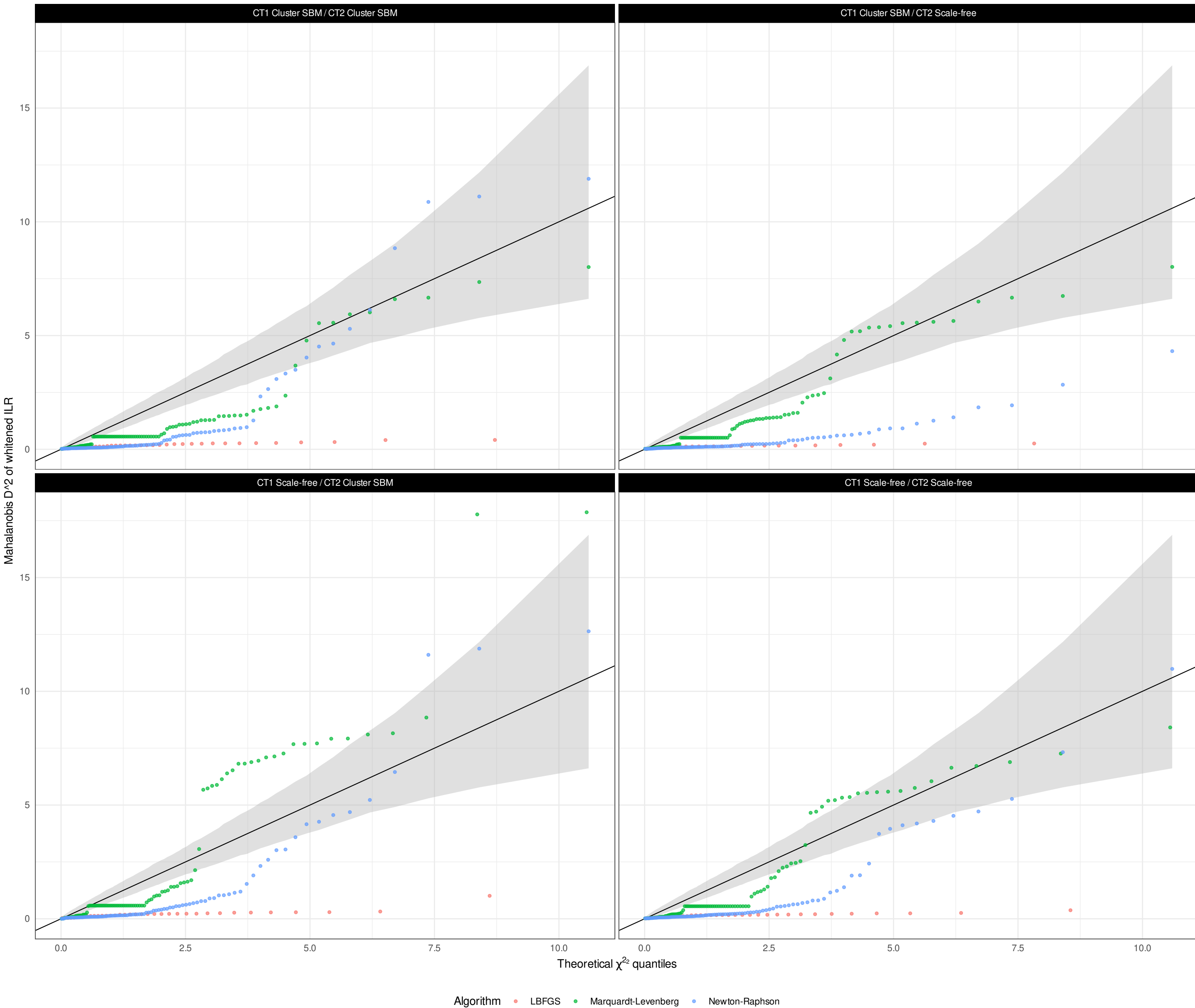
$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.

moderately unbalanced / high OVL



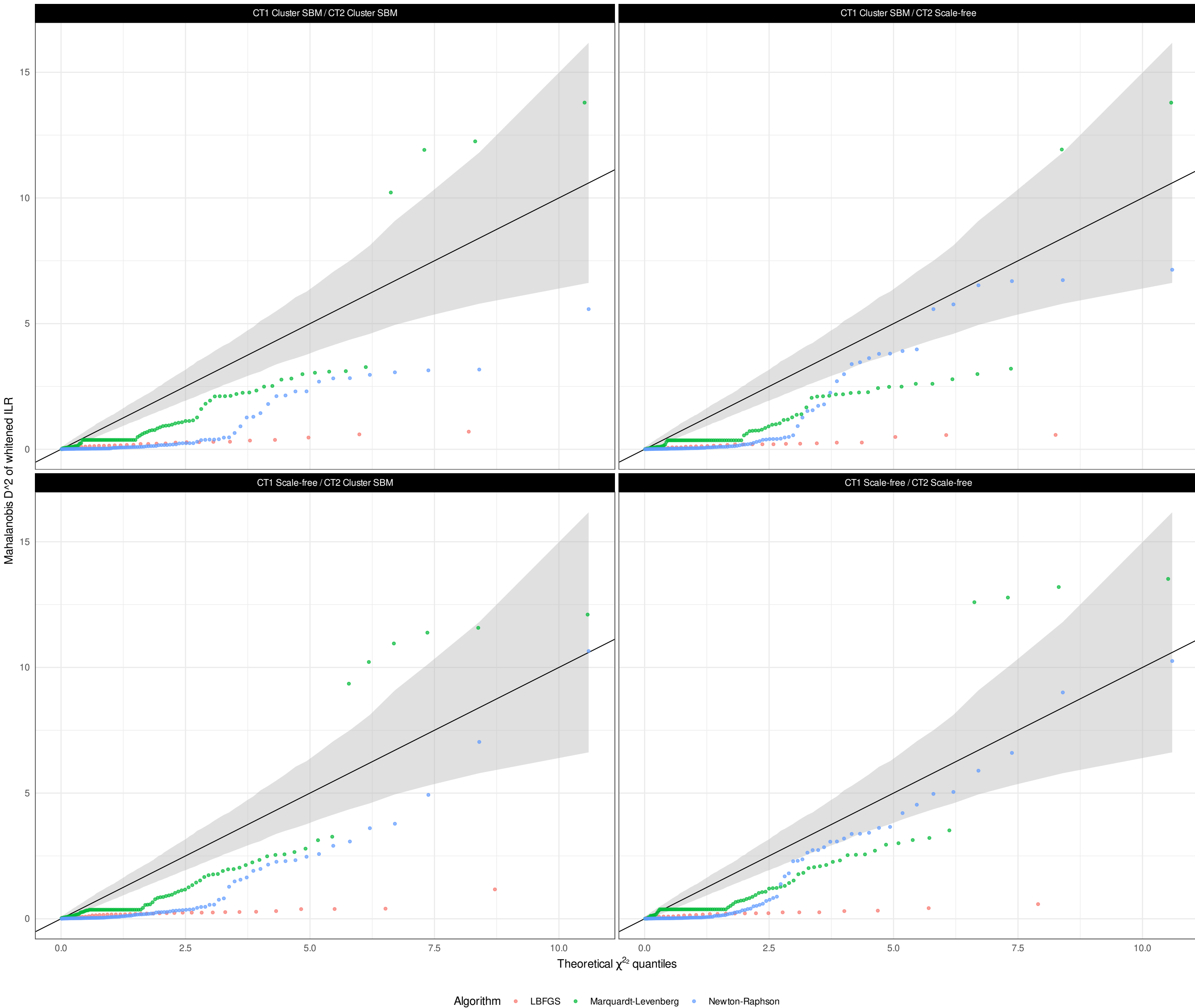
$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.

highly unbalanced / low OVL



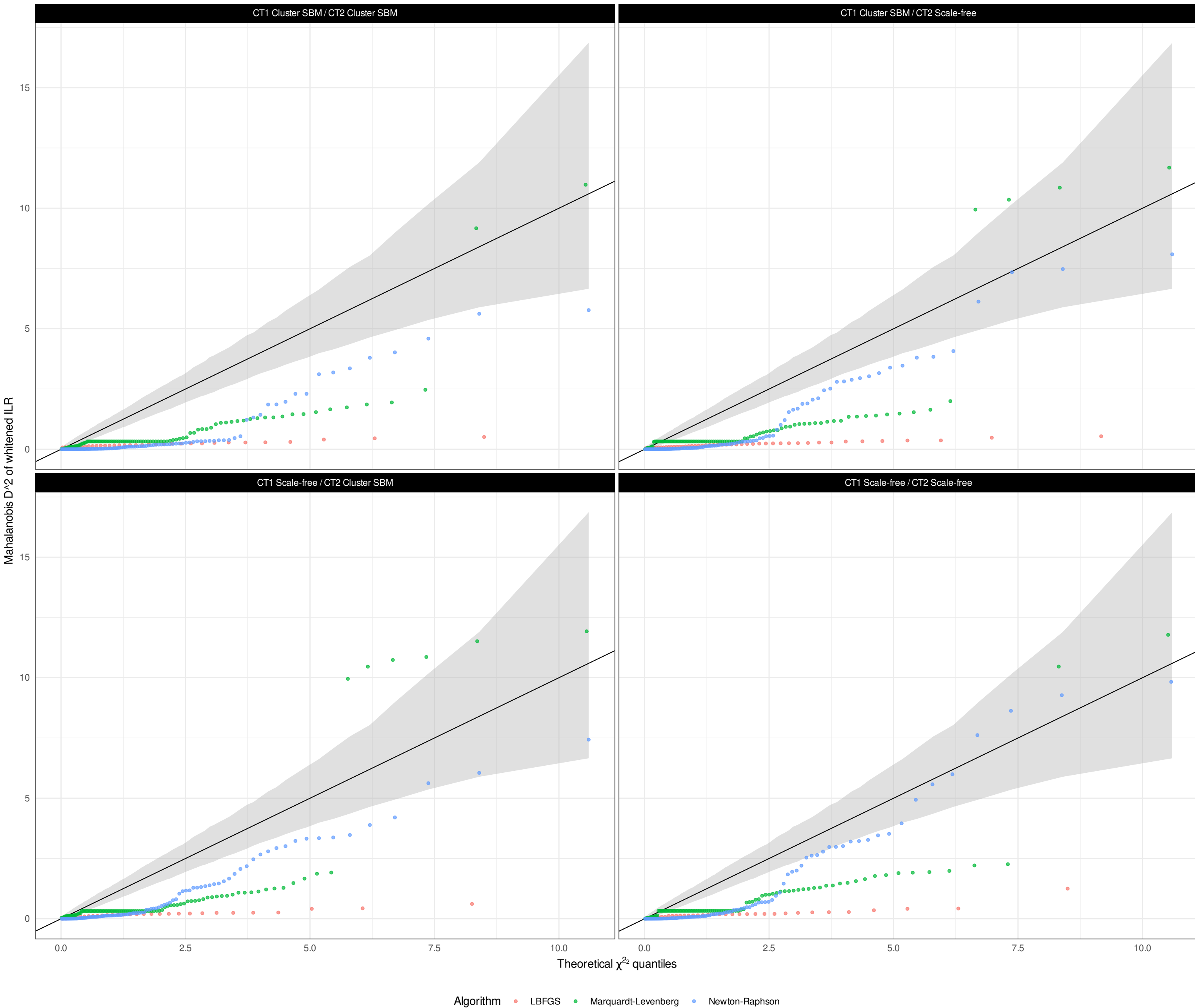
$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.

highly unbalanced / moderate OVL



$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.

highly unbalanced / high OVL



$D^2 = W^T W$ versus χ^2_{J-1} (joint covariance of the ILR MLE). Grey band: 95% Monte Carlo envelope under the exact χ^2 null. LBFGS, Marquardt-Levenberg and Newton-Raphson; identity reference.